

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 9: Singular Value Decomposition (SVD)

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1 Keywords

- Singular Value Decomposition, SVD, general linear maps

2 Main Points

2.1 Singular Value Decomposition

- Any $m \times p$ matrix, $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ (for a design matrix, m is the number of data points and p the number of features)
 - $\mathbf{U}$ is an $m \times m$ orthogonal matrix
 - $\mathbf{S}$ is an $m \times p$ matrix with zeros everywhere except the diagonal where $S_{ii} = s_i \geq 0$
 - $\mathbf{V}$ is a $p \times p$ orthogonal matrix
- The values s_i are known as the *singular values* of $\mathbf{X}$
- The SVD of a symmetric matrix is just the eigen-decomposition
- **Economical SVD**
 - If $m > p$ some algorithms won't bother outputting the last $m - p$ columns of $\mathbf{U}$
 - If $p > m$ some algorithms won't bother outputting the last $p - m$ columns of $\mathbf{V}$
 - In this case it will output a square $\mathbf{S}$ matrix

2.2 General Linear Mapping

- Recall that matrices are the most general linear operators
- Since any matrix $\mathbf{M}$ can be written as $\mathbf{U} \mathbf{S} \mathbf{V}^T$ we can interpret any linear mapping as doing three operations
 - A rotation (with possibly a reflection) defined by $\mathbf{V}^T$
 - A rescaling of each coordinate by s_i
 - A rotation (with possibly a reflection) defined by $\mathbf{U}$
- **Duality**
 - Using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then
 - $\mathbf{C} = \mathbf{X} \mathbf{X}^T = \mathbf{U} \mathbf{S} \mathbf{S}^T \mathbf{U}^T$
 - $\mathbf{D} = \mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T$
 - $\mathbf{S} \mathbf{S}^T$ and $\mathbf{S}^T \mathbf{S}$ are diagonal matrices whose non-zero diagonal elements are s_i^2

2.3 Ridge Regression

- Ridge regression is linear regression with an L_2 regulariser
- Adding a regulariser $\eta \|\mathbf{w}\|^2$ to the squared loss, the weights, $\mathbf{w}^*$, that minimise the loss function are given by $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$
- Using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then

$$\mathbf{w}^* = \mathbf{V} \hat{\mathbf{S}}^+ \mathbf{U}^T \mathbf{y}$$

where $\hat{\mathbf{S}}^+$ is a regularised pseudo-inverse of $\mathbf{S}$ given by

$$\hat{\mathbf{S}}^+ = (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^T$$

- If $\eta = 0$ (and all $s_i > 0$) this is equal to the pseudo-inverse of $\mathbf{S}$
- $\hat{\mathbf{S}}^+$ is a $p \times m$ matrix which is zero everywhere except on the diagonal, where $\hat{S}_{ii}^+ = \frac{s_i}{s_i^2 + \eta}$
 - Note if $s_i = 0$ linear regression has an infinity of solutions and the pseudo-inverse of $\mathbf{X}$, as we derived it, does not exist (setting $\eta = 0$ we get $S_{ii}^+ = 1/s_i$ which is not defined when $s_i = 0$). See the Moore–Penrose pseudo-inverse below
 - In the regularised case $\hat{S}_{ii}^+ = 0$ (we have selected one of the solutions that minimise the squared error)
 - If $s_i \ll \eta$ then without the regularisation term the inverse is very ill-conditioned while with the regularisation term $\hat{S}_{ii}^+$ will be small
 - If $s_i \gg \eta$ then $\hat{S}_{ii}^+ \approx \frac{1}{s_i} = S_{ii}^+$
- Adding an L_2 regulariser means that the optimum weights, $\mathbf{w}^*$, will be less sensitive to the training data reducing the variance in the bias-variance dilemma

2.4 The Moore–Penrose Pseudo-Inverse (going further)

- The pseudo-inverse we derived, $\mathbf{X}^+ = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$, only exists when $\mathbf{X}^T \mathbf{X}$ is invertible. That needs every one of the p singular values to be non-zero, which is impossible when $m < p$ (the under-constrained case)
- Using the SVD it can be written $\mathbf{X}^+ = \mathbf{V} \mathbf{S}^+ \mathbf{U}^T$, where $\mathbf{S}^+$ is the $p \times m$ diagonal matrix with $S_{ii}^+ = 1/s_i$
- The *Moore–Penrose pseudo-inverse* makes one change: wherever $s_i = 0$ it sets $S_{ii}^+ = 0$ instead of $1/s_i$

$$S_{ii}^+ = \begin{cases} 1/s_i & s_i \neq 0 \\ 0 & s_i = 0 \end{cases}$$

This always exists. When $\mathbf{X}^T \mathbf{X}$ is invertible it is the same as our pseudo-inverse, and when $\mathbf{X}$ is square and invertible it is $\mathbf{X}^{-1}$

- $\mathbf{w} = \mathbf{X}^+ \mathbf{y}$ minimises $\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2$ and, of all the weight vectors that do so, it is the one with the smallest $\|\mathbf{w}\|$
- To see why, rotate to the coordinates given by the SVD: let $\mathbf{z} = \mathbf{V}^T \mathbf{w}$ and $\mathbf{c} = \mathbf{U}^T \mathbf{y}$. Orthogonal matrices do not change lengths, so $\|\mathbf{w}\| = \|\mathbf{z}\|$ and

$$\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2 = \|\mathbf{U}^T (\mathbf{U} \mathbf{S} \mathbf{V}^T \mathbf{w} - \mathbf{y})\|^2 = \|\mathbf{S} \mathbf{z} - \mathbf{c}\|^2 = \sum_{i: s_i \neq 0} (s_i z_i - c_i)^2 + \sum_{i: s_i = 0} c_i^2$$

(in the second sum i runs over the rows of $\mathbf{S}$ that have no non-zero singular value)

- The loss is minimised by $z_i = c_i/s_i$ whenever $s_i \neq 0$
- The other components of $\mathbf{z}$ do not appear in the loss at all: these are the directions $\mathbf{v}_i$ with $\mathbf{X}\mathbf{v}_i = \mathbf{0}$, where the data give no information
- Every choice of them fits equally well. Setting them to zero gives the smallest $\|\mathbf{z}\| = \|\mathbf{w}\|$, and that is exactly what $S_{ii}^+ = 0$ does
- It is also the limit of ridge regression as the regularisation vanishes: as $\eta \rightarrow 0$, $\hat{S}_{ii}^+ = s_i/(s_i^2 + \eta)$ tends to $1/s_i$ if $s_i \neq 0$ and is 0 if $s_i = 0$
- This is what `np.linalg.pinv` computes, and `np.linalg.lstsq` returns the same minimum-norm solution. In practice "zero" means below a small tolerance, so tiny singular values caused by rounding errors are discarded rather than inverted

3 Exercises

3.1 Ridge regression

- Ridge regression is just linear regression with an L_2 regulariser
 1. Derive the optimal weights in ridge regression
 2. Show that using $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ then $\mathbf{w}^* = \mathbf{V}(\mathbf{S}^T\mathbf{S} + \eta\mathbf{I})^{-1}\mathbf{S}^T\mathbf{U}^T\mathbf{y}$
- See answers

4 Experiments

4.1 SVD

- Using Matlab/Octave or python have a play with `svd`

```
X = randn(3,4)           % construct a random matrix
[U,S,V] = svd(X)         % compute singular value decomposition
U*S*V'                   % should be the same as X
U*U'                     % should be the identity up to round error
U'*U                     % should be the identity up to round error
V*V'                     % should be the identity up to round error
V'*V                     % should be the identity up to round error
[Ue,L1] = eig(X*X')      % Ue should be the same as U up to permutation
S*S'                     % same as L1 up to permutation
[Ve,L2] = eig(X'*X)      % Ve should be the same as V up to permutation
S'*S                     % same as L2 up to permutation
```

```
inv(X'*X + 0.1*eye(4))    % check identity
V*inv(S'*S + 0.1*eye(4))*V' % should be the same
```

4.2 Verify Identity

- Again use Matlab/Octave or python
- For a random 4×5 matrix $\mathbf{X}$
 - Check that using $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ that

$$(\mathbf{X}^T\mathbf{X} + \eta\mathbf{I})^{-1} = \mathbf{V}(\mathbf{S}^T\mathbf{S} + \eta\mathbf{I})^{-1}\mathbf{V}^T$$

holds for some random matrix using Matlab/Octave or python

- Examine $\mathbf{S}^T \mathbf{S}$, $\mathbf{S}^T \mathbf{S} + 0.1 \mathbf{I}$, $(\mathbf{S}^T \mathbf{S} + 0.1 \mathbf{I})^{-1}$ and $(\mathbf{S}^T \mathbf{S} + 0.1 \mathbf{I})^{-1} \mathbf{S}^T$
- See if you can invert $\mathbf{X}^T \mathbf{X}$: it is singular, but due to rounding errors it might be inverted (it was a scary matrix when I tried it)

```
X = randn(4,5)           % construct a random matrix
[U,S,V] = svd(X)         % compute singular value decomposition

inv(X'*X + 0.1*eye(5))   % check identity
V*inv(S'*S + 0.1*eye(5))*V' % should be the same

S'*S                     % singular
S'*S + 0.1*eye(5)        % now invertible
inv(S'*S + 0.1*eye(5))   % 
inv(S'*S + 0.1*eye(5))*S' % 5x4 diagonal matrix

inv(X'*X)                 % shouldn't be able to do this
```

5 Answers

5.1 Ridge regression

1. It is straightforward to show

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

2. The only hard part is to show that

$$(\mathbf{X}^T \mathbf{X} + \eta \mathbf{I})^{-1} = \mathbf{V} (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{V}^T$$

- It is easy to show that $\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T$
- But we also have $\mathbf{I} = \mathbf{V} \mathbf{V}^T$ as $\mathbf{V}$ is an orthogonal matrix
- Thus $\mathbf{M} = \mathbf{X}^T \mathbf{X} + \eta \mathbf{I} = \mathbf{V} (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I}) \mathbf{V}^T = \mathbf{V} \mathbf{W} \mathbf{V}^T$ where $\mathbf{W} = \mathbf{S}^T \mathbf{S} + \eta \mathbf{I}$
- But $(\mathbf{A} \mathbf{B} \mathbf{C})^{-1} = \mathbf{C}^{-1} \mathbf{B}^{-1} \mathbf{A}^{-1}$ (which we can verify by multiplying $\mathbf{C}^{-1} \mathbf{B}^{-1} \mathbf{A}^{-1}$ on either the left or right by $\mathbf{A} \mathbf{B} \mathbf{C}$)
- Thus $\mathbf{M}^{-1} = (\mathbf{V} \mathbf{W} \mathbf{V}^T)^{-1} = (\mathbf{V}^T)^{-1} \mathbf{W}^{-1} \mathbf{V}^{-1} = \mathbf{V} \mathbf{W}^{-1} \mathbf{V}^T$ where we use $\mathbf{V}^{-1} = \mathbf{V}^T$ as $\mathbf{V}$ is an orthogonal matrix